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* $T = \text{estimate of } \theta$, $\theta = \text{parameter}$.
 T is calculated from samples $X_1, \dots, X_n$ drawn from a dist.ⁿ

$$E_\theta[(T-\theta)^2] = R(\theta, T) = \text{MSE/Risk}$$

• T_1 is better than T_2 when,

$$R(\theta, T_1) \leq R(\theta, T_2) \quad \forall \theta \in \mathcal{H} \rightarrow \text{parameter space}$$

Sample = X (Observation)

$$\text{Let } X \sim \text{Bin}(\theta, n), \quad 0 < \theta < 1$$

Q) Can there be a T^* which is supposedly the "best" estimator of θ ?

A) NO.

If T^* is the "best" estimator, then $R(\theta, T^*) \leq R(\theta, T) \quad \forall \theta, T$

Let $T(X) = \theta_0$ for some $\theta_0 \in (0, 1)$.

$$\therefore R(\theta_0, T^*) \leq R(\theta_0, T) = R(\theta_0, \theta_0) = 0$$

$$\Rightarrow E_{\theta_0}[(T^* - \theta_0)^2] = 0$$

$$\Rightarrow T^* = \theta_0$$

However, when $\theta = \frac{\theta_0}{2}$, T^* won't be the "best" estimator.

$\therefore$ There are no "best" estimators in Ra-sing-se if we consider this ~~me~~ metric.

* $X = \text{Sample / Data / Observation}$

$\hookrightarrow$ Random w/ a prob. dist.ⁿ

$$\mathcal{P} \in \mathcal{P} = \{P_\theta \mid \theta \in \mathcal{H}\}, \quad \mathcal{H} \rightarrow \text{parameter space}$$

$T = \text{estimate / statistic} = T(X)$

$$R(\theta, T) \rightarrow \text{Risk function} = E_\theta[(T-\theta)^2]$$

$$\text{Bias}(T) = E_\theta(T - \theta)$$

$$E_{\theta}[(T-\theta)^2] = [E_{\theta}(T)-\theta]^2 + \text{Var}_{\theta}(T) = \text{Bias}(T)^2 + \text{Var}_{\theta}(T)$$

(Why? $[E_{\theta}(T)-\theta]^2 + E_{\theta}[(T-E_{\theta}(T))^2]$ then proceed)

- When $\text{Bias}(T) = 0 \forall \theta$, we call T an unbiased estimate.

* A question: For an unbiased estimate T , $E_{\theta}(T) = \theta \forall \theta$

Note: Unbiased estimators aren't unique.

i) - Can we get a "best" unbiased estimate? (If such an estimate exists, it's called the Uniformly Minimum Variance Unbiased Estimator $\rightarrow$ UMVUE)

now) Example 1: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bin}(1, \theta)$, $0 < \theta < 1$

$$X = (X_1, \dots, X_n)$$

$$\text{Let } T^*(X) = \frac{X_1 + \dots + X_n}{n} = \bar{X}$$

In Ex 10, T^* is an unbiased estimator of θ that attains the Cramer-Rao lower bound. Hence, it is a / the UMVUE, which is $\frac{\theta(1-\theta)}{n}$.

ii) - If we do have an UMVUE, will it be unique?

A) Let T_1 & T_2 be 2 UMVUE's for θ

Then, $T = \frac{T_1 + T_2}{2}$ is also an unbiased estimator.

We know, $\text{Var}(T_1) = \text{Var}(T_2) = \sigma^2$ (say).

Let $\text{Cov}(T_1, T_2) = \rho \in [-1, 1]$

Now, we see: $\sigma^2 \leq \text{Var}(T)$

$$\Rightarrow \sigma^2 \leq \frac{1}{4} (\sigma^2 + \sigma^2 + 2\rho\sigma^2)$$

$$\Rightarrow 1 \leq \rho \Rightarrow \rho = 1 \Rightarrow T_1 = T_2$$

$\therefore$ Result: When UMVUE ~~exists~~ exists it is ~~unique~~ unique.

Example 2: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$

$$X = (X_1, \dots, X_n)$$

An unbiased estimator: $T(X) = \bar{X}$

$$\therefore \text{CR-lower bound} = \frac{1}{n} = \text{Var}_\theta(\bar{X})$$

$\therefore T(X)$ is UMVUE & its unique.

Example 3: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$

Unknown $\rightarrow \theta \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+$

$$T_\theta = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \rightarrow \text{Unbiased estimator for } \sigma^2$$

$$T' = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \rightarrow \text{Not unbiased but lesser risk}$$

Let $T_c = c \sum_{i=1}^n (X_i - \bar{X})^2$. For what c , T_c will have min. risk?

$$\begin{aligned} R(\sigma^2, T_c) &= [E_\theta(T_c) - \sigma^2]^2 + c^2 \text{Var}_\theta\left(\sum_{i=1}^n (X_i - \bar{X})^2\right) \\ &= [c\sigma^2(n-1) - \sigma^2]^2 + c^2 \sigma^4 2(n-1) \\ &= c^2 \sigma^4 (n-1)^2 + \sigma^4 - 2c\sigma^4(n-1) + c^2 \sigma^4 2(n-1) \\ &= [c^2(n-1)^2 + 2(n-1) - 2c(n-1) + 1] \sigma^4 \end{aligned}$$

$$\text{For min. : } c(n^2 - 2n + 1 + 2n - 2) - 2n + 2 = 0$$

$$\Rightarrow c(n^2 - 1) = 2 - 2n$$

$$\Rightarrow c = \frac{2(1-n)}{(n-1)(n+1)} = \frac{-2}{n+1}$$

For min. : $(c^2(n^2-1) - 2c(n-1) + 1)$ ~~@@@~~
 $\Rightarrow c = \frac{1}{n+1}$ # (somehow)

Example: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U[0, \theta]$

$$X = (X_1, \dots, X_n), \theta \in \mathbb{R}_+$$

2 unbiased estimators for θ

$$T_1 = 2\bar{X}, \quad T_2 = \frac{n+1}{n} X_{(n)}$$

$$\therefore \text{Risk}(\theta, T_1) = \frac{\theta^2}{3n}, \quad \text{Risk}(\theta, T_2) = \textcircled{\#}$$